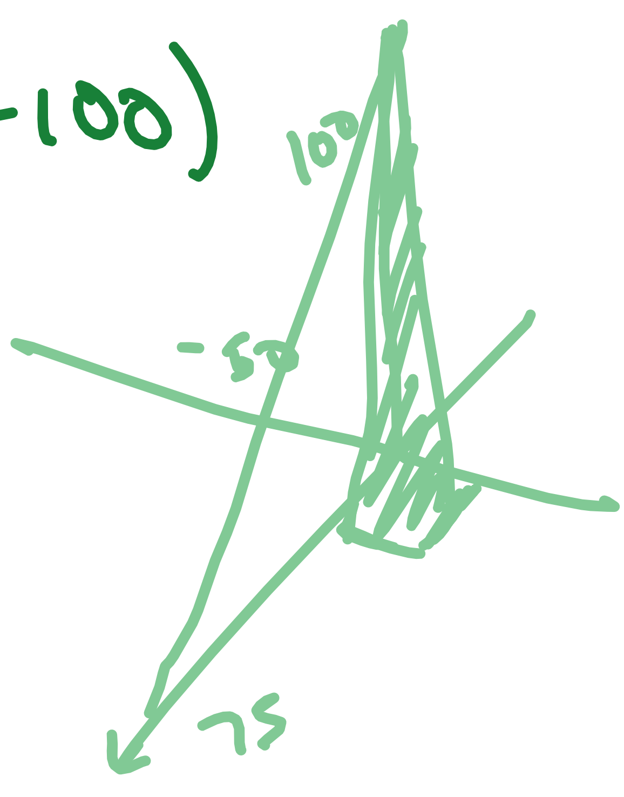
Application of Vectors

Section 11.2

8-17-26

A) 3D APPLICATION

ex #100)



$a\langle 75, -50, -100\rangle$

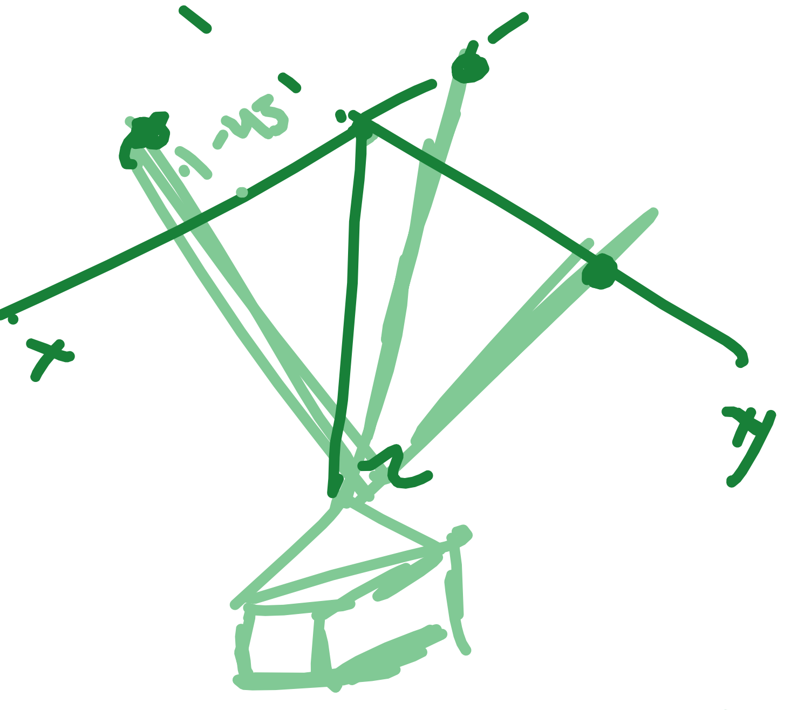
$25a\langle 3, -2, -4\rangle$

$25a\sqrt{29} = 550$

$a = \frac{22}{\sqrt{29}}$

$\frac{22}{\sqrt{29}}\langle 75, -50, -100\rangle$

ex #103)



$\overrightarrow{AB} = a\overset{\langle 0, 70a, 115a\rangle}{\langle 0, 70, 115\rangle}$

$\overrightarrow{AC} = b\overset{\langle -60b, 0, 115b\rangle}{\langle -60, 0, 115\rangle}$

$\overrightarrow{AD} = c\overset{\langle -45c, -65c, 115c\rangle}{\langle -45, -65, 115\rangle}$

$$\begin{array}{c c} & \begin{array}{ccc} a & b & c \end{array} \\ \begin{array}{c} x \\ y \\ \\ \end{array} & \begin{bmatrix} 0 & -60 & 45 \\ 70 & 0 & -65 \\ 115 & 115 & 115 \end{bmatrix} \end{array} \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

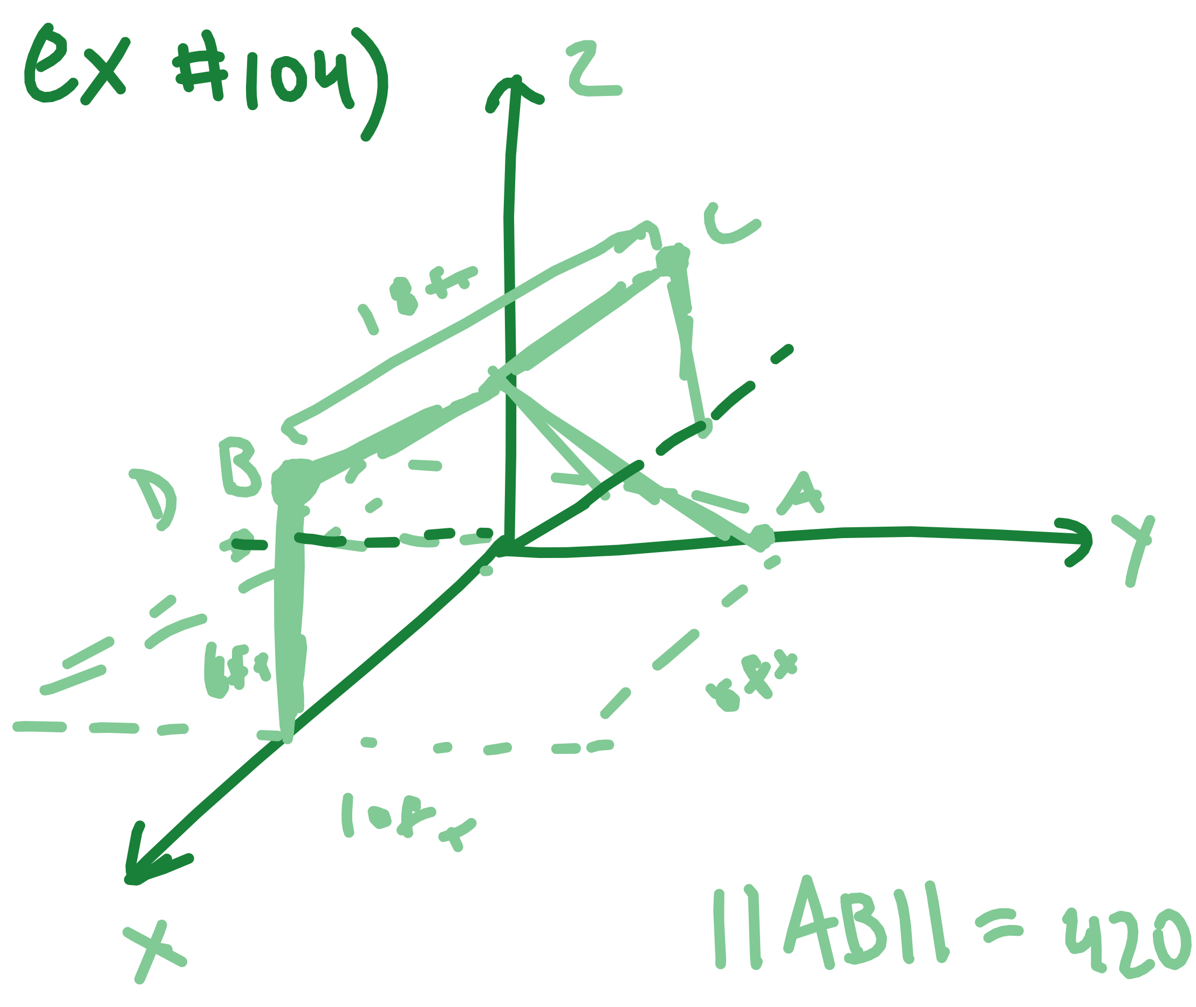
$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 1.507 \\ 1.217 \\ 1.623 \end{bmatrix}$$

$\|\overrightarrow{AB}\| = 1.507 \cdot \sqrt{18125} = 202.866$

$\|\overrightarrow{AC}\| = 1.217 \cdot \sqrt{16825} = 157.859$

$\|\overrightarrow{AD}\| = 1.623 \cdot \sqrt{19475} = 226.494$

ex #104)



$A(0, 10, 0)$

$B(8, 0, 6)$

$C(-10, 0, 6)$

$\|AB\| = 420$

$\|AC\| = 650$

$\overrightarrow{AB} = a\langle 8, -10, 6\rangle$

$\overrightarrow{AC} = b\langle -10, -10, 6\rangle$

$2a\sqrt{50} = 420$

$2b\sqrt{59} = 650$

$a = \frac{210}{\sqrt{50}}$

$b = \frac{325}{\sqrt{59}}$

$\overrightarrow{AB} + \overrightarrow{AC} = \langle 8a - 10b, -10a - 10b, 6a + 6b\rangle$

$\|\overrightarrow{AB} + \overrightarrow{AC}\| = 860.022$